

PROJECT DOCUMENTATION



Title: DAY OF THE WEEK FINDER

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1. Problem Definition

The problem definition includes the brief introduction about the project, technology used, and the field of the project.

- **Current Challenge:** Manually calculating the exact day of the week for a random historical or future date is complex and prone to mathematical errors.
- **Proposed Solution:** A Python-based GUI application that takes a date as input, automatically handles leap year exceptions, and instantly computes the correct day of the week.
- **Technology Stack:** Python (Core Logic) and Tkinter (Graphical User Interface).

2. Objective

This describes the objective of the desired project.

- **Accurate Calculation:** To implement a mathematical algorithm (similar to Zeller's congruence) with month and year codes to find the day of the week.
- **Data Validation:** To strictly validate user input in DD/MM/YYYY format and reject invalid dates (e.g., 30/02/2024 or 31/04/2023).
- **User Experience:** To provide a clean, accessible interface using Tkinter where users can easily input data, press Enter, and see formatted results.

3. Feasibility Study

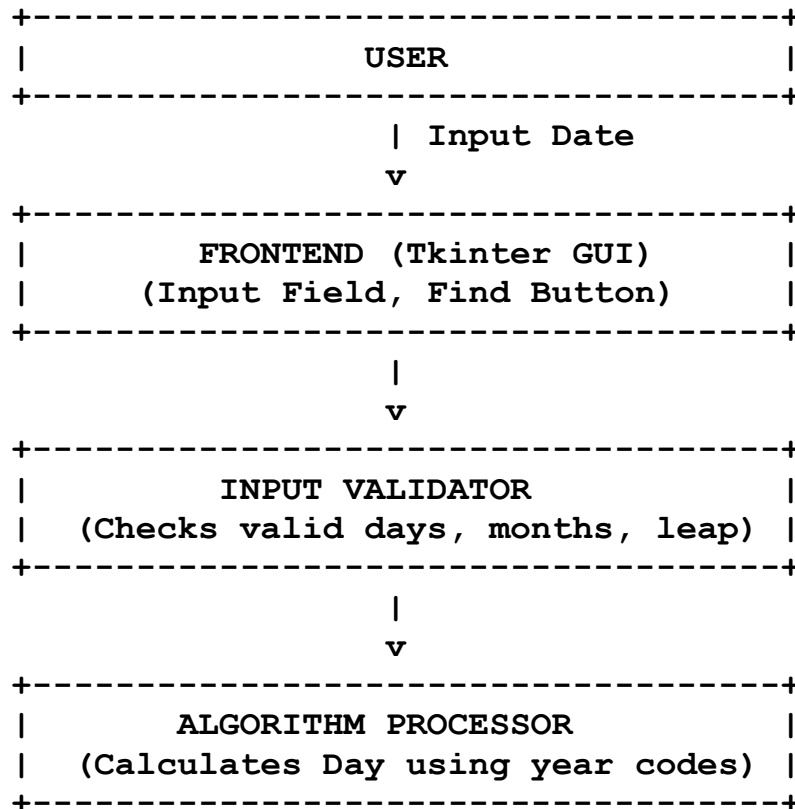
This describes the feasibility, need and significance of the project.

- **Technical Feasibility:** The project uses standard mathematical operators and Python's built-in Tkinter library, requiring no external dependencies.
- **Operational Feasibility:** Very high, as the user only needs to type a date and click a button. The system handles all backend complexity.

4. Architectural Design

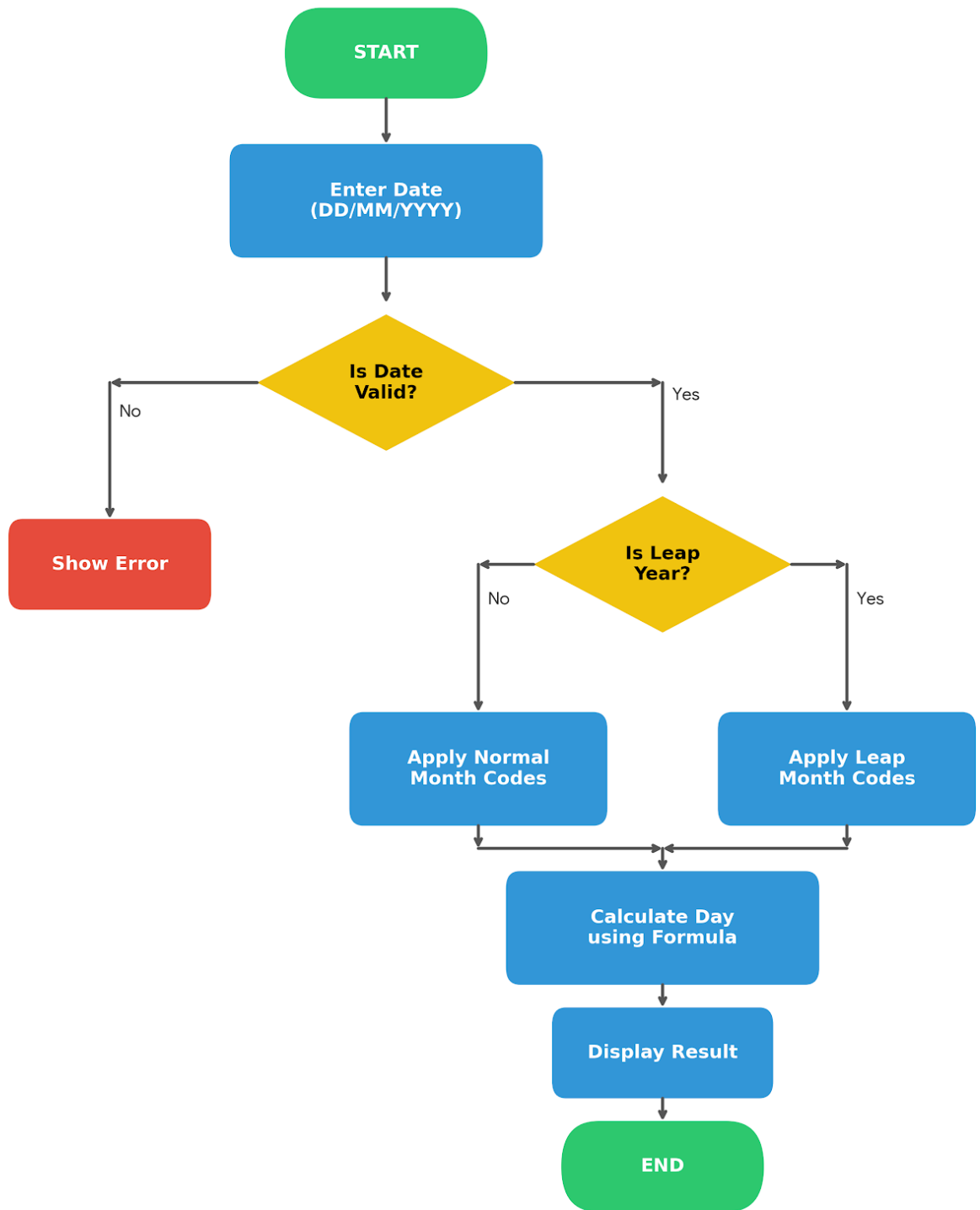
This section illustrates the structural and behavioral design of the Day Finder System.

The diagram below shows how user input flows from the graphical interface to the validation and algorithmic processing units.



4.1 Flow Chart

The following flow chart demonstrates the decision-making process for leap years and data validation.



5. Methodology

Steps followed to achieve the objective of the Project.

- **Requirement Analysis:** Determining the mathematical rules for leap years (divisible by 4, but century years must be divisible by 400).
- **Algorithm Design:** Formulating arrays for month_code, year_code, and day_name to compute the final index.
- **Implementation:** Coding the GUI and binding the <Return> key for quicker execution.
- **Exception Handling:** Implementing try-except blocks and conditional checks to raise ValueError for impossible dates.

6. Core Code Implementation

The core mathematical logic to calculate the day of the week is based on specific century and month codes.

```
def leap_year_day(day, month, year):  
    day_count=0  
    # Arrays for names and codes  
    month_name =  
    ["January", "February", "March", "April", "May", "June", "July",  
  
    "August", "September", "October", "November", "December"]  
  
    day_name=['Saturday', 'Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday']  
    month_code=[0, 3, 4, 0, 2, 5, 0, 3, 6, 1, 4, 6] # Leap  
    year codes  
    year_code=[6, 4, 2, 0]
```

```

last_two_digit_of_year = year % 100
rem = (year // 100) % 4

# Core Mathematical Formula
day_count += ((day % 7) +
month_code[month-1] + year_code[rem] +
              (last_two_digit_of_year % 7) +
              (last_two_digit_of_year // 4)) % 7

# Formatting Result...
result_text = ( f"Day:
{day_name[day_count]}\n"
               f>Date:
{day:02d}/{month:02d}/{year}\n"
               f"Leap Year:  Yes")
result_label.config(text=result_text,
fg="green")

```

7. Tools & Platform Used

Details of the tools and platform used.

- **Programming Language:** Python 3.x
- **Graphical Interface:** Tkinter (Python Standard Library)
- **IDE:** VS Code / PyCharm / IDLE

8. References

Here specify the description of the study material, papers, reports etc. referred for the development of the project.

- **Python Official Documentation:** For try-except exception handling and integer mapping.

- **Tkinter Documentation:** For GUI widget configurations and event binding (e.g., <Return> key integration).